Homework assignment for 02YMECH
Chapter: Kinematics

1. A particle is under uniform circular motion (counter-clockwise) with a radius of 5 m and a constant speed of 10 m/s. The center of the circle is at point (3m, 4m) in the xy-plane.

   (a) Write the position vector of the particle as a function of time, assuming it starts at the point (8m, 4m) at time t = 0.

   (b) Calculate the velocity and acceleration vectors at $t = \pi/2$ seconds and show that they are perpendicular to each other by considering their dot product. Is this true for all time?

2. The position vectors of two particles are given by:

$$\mathbf{r}_1(t) = (1 + 2t)\,\hat{\imath} + (3 - t)\,\hat{\jmath}$$

$$\mathbf{r}_2(t) = (4 - t)\,\hat{\imath} + (1 + 3t)\,\hat{\jmath}$$

   (a) Find the velocity and acceleration of each particle.

   (b) Find the time at which the two particles are closest to each other and determine that minimum distance.